Joe Crop

Lake Oswego, OR • joe.crop@gmail.com • (503) 332-4445
https://joecrop.com • https://linkedin.com/in/joecrop

SUMMARY

IC architect, algorithm designer, and serial entrepreneur with 15+ years spanning silicon tape-out, mixed-signal design, and company building. PhD in ECE from Oregon State University. 30+ tape-outs across consumer, automotive, medical, and energy applications.

EDUCATION

PhD, Electrical & Computer Engineering — Oregon State University, 2014
Dissertation: Resilient ultra-low-power digital mixed-signal ICs — reliability under near-threshold operation

BS, Electrical & Computer Engineering — Oregon State University, 2009 — Minor: CS

EXPERIENCE

Co-Founder & SVP, Digital & Algorithms — Overlord Labs — Jul 2025–Present
Building the next-generation BMS SoC (Overlord Genesis™): embedded RISC-V core, physics-based on-chip electrochemical battery modeling, autonomous protection, and fully programmable OEM firmware. Leading digital architecture, microprocessor design, and battery algorithm development.

CEO & Founder — LTIAAS — Mar 2021–Present
Founded and scaled a B2B SaaS API platform simplifying LTI (Learning Tools Interoperability) for edtech developers and publishers. Scaled to 1.5M+ monthly API requests across every continent.

Senior Principal Engineer — Integense Micro — Sep 2022–Jun 2025
Advanced mixed-signal IC design and architecture for high-performance signal chain ICs. Led digital RTL, verification, and top-level integration.

Advisor & Principal Engineer — Xceltronics / CRX Microtech — Aug 2022–Jun 2025
Led digital design for the CRX1 (16S BMS monitor/protector/balancer IC) and CRX0 (single-cell Li-ion protector). Defined architecture, RTL, and verification strategy.

Principal Member of Technical Staff — Analog Devices (f.k.a. Maxim Integrated)
2014–2022

- *IP Design & Verification:* Pipelined and SAR ADCs, Hybrid PLLs, power converters, memory controllers, RISC-V mixed-signal blocks, OTP, I2C, SPI, and process qualification chips.
- *Products:* RF-to-bits Hybrid-Radio Receivers (MAX2175), Audio DACs, Ultrasound front-ends, Battery Management ICs (MAX77865S, MAX11261), Power Converters.
- *CAD / Automation:* Architected an automated chip generator covering AMS+UVM testbench and regmap/model generation plus reusable mixed-signal IP; deployed across most new Maxim design starts.

I/O Design Intern — Intel — Summer 2012
I/O circuits for next-generation 22nm server processors; designed novel circuits and verified operation across PVT corners.

Instructor & Research Assistant — Oregon State University 2009–2014

Taught ECE 473/573 (digital design, SystemVerilog, synthesis). Researched resilient ultra-low-power VLSI and bio-medical sensor SoCs (SRC, NSF, DOE, DARPA funded). Research intern at Fudan University, Shanghai (Summer 2011).

TECHNICAL SKILLS

IC Design & EDA: SystemVerilog, Verilog-AMS, VHDL, UVM, SPICE/Spectre, Virtuoso, Synopsys Design Vision / PrimeTime, Cadence Xcelium / SimVision / SKILL / OCEAN, Calibre (DRC/LVS), SoC Encounter (CPF/UPF), Xilinx FPGA, Altium / KiCad, MATLAB, PBS / LSF / SGE

Languages & Frameworks: Python, C/C++, JavaScript/TypeScript, React, Vue.js, Next.js, Express.js, Node.js, PHP, Java, Perl, Bash/Shell, SQL, GraphQL, REST APIs, WebSockets

DevOps & Cloud: Git, Perforce, Docker, Kubernetes, AWS, Linux/Ubuntu, PostgreSQL, MongoDB, MySQL, Redis, Firebase, Nginx, GitHub Actions, GitLab CI, Netlify

SELECTED PUBLICATIONS

J. Crop et al., “A Testbench Creation and Maintenance Tool To Enhance IP Reuse,” GTC, 2022

J. Crop et al., “A Verification-Driven Model for IP Delivery,” Functional Verification Summit, 2021

B. Querbach . . . **J. Crop**, P.Y. Chiang, “Architecture of a Reusable BIST Engine . . . on 14-nm SoC,” IEEE Design & Test, 2016

R. Pawlowski, **J. Crop** et al., “Characterization of Radiation-Induced Soft Errors from 0.33V to 1.0V in 65nm CMOS,” CICC 2014

Crop, J.; Pawlowski, R.; Chiang, P., “Regaining Throughput Using Completion Detection for Error-Resilient, Near-Threshold Logic,” DAC 2012

Pawlowski . . . **Crop, J.** et al., “Synctium-I: A 530mV, 10-Lane SIMD Processor with Variation Resiliency in 45nm-SOI,” ISSCC 2012

Crop, J. et al., “150mV Sub-Threshold Asynchronous Multiplier for Low-Power Sensor Applications,” VLSI-DAT 2010